

Amazon Order Review Analysis for 2022 and 2023

By: Taylor Martin

February 26, 2026

Section 1. Introduction

This custom data visualization project consists of a clean dataset and six specific goals and corresponding visuals created to generate narratives to lead to potential business insights. The dataset, storyboard and completed report pdf are all housed publicly on Github at: <https://github.com/tmartin-m/a7-final-project>

Section 2. Data Description

The dataset is a subset of retail/e-commerce transactions for Amazon spanning 2022 and 2023 across North America, Europe, Asia and the Middle East. The dataset includes transaction sales data including product sales, customer purchasing behavior, pricing, discounts and total revenue. The data set could be used for forecasting future sales, current trend analysis or for business intelligence decisions.

The dataset is tabular with each row representing 1 completed transaction and each column representing 1 of 13 specific attributes for the transaction. The attributes included are listed below:

- order_id: a unique number identifier assigned to each order
- order_date: the date the order was placed
- product_id: a unique number identifier for the product purchased in the transaction
- product_category: the category of the product purchased in the transaction (i.e. Beauty, Books, Electronics, Fashion, Home & Kitchen or Sports)
- price: the original price of the product before discounts
- discount_percent: discount percentage applied to the transaction
- quantity_sold: number of units sold of the product in the transaction
- customer_region: Country/Continent the customer ordered in (i.e. North America, Europe, Middle East or Asia)
- payment_method: the method of payment used in the transaction (i.e. Cash on Delivery, Credit Card, Debit Card, UPI, or Wallet)
- rating: customer rating for the product on a scale of 1 to 5
- review_count: number of reviews for the product
- discount_price: price of the product after the discount percentage is applied
- total_revenue: total price of the transaction for all products purchased after the discount is applied

Of the attributes 8 are numerical including order_date, price, discount_percent, quantity_sold, rating, review_count, discounted_price and total_revenue. While 3 are categorical: product_category, customer_region and payment_method. The data set consists of 13 columns and 50,000 rows of orders. The dataset was downloaded from a Kaggle Url: <https://www.kaggle.com/datasets/aliihussain/amazon-sales-dataset>.

Taylor Martin

February 26, 2026

Section 3. Data Cleaning Strategies

No data cleaning was required for this dataset. It was downloaded clean, well structured and ready for analysis.

Section 4. Clean Dataset –

The dataset covers 50,000 Amazon customer transactions covering both 2022 and 2023 across 6 different product categories, 5 different payment methods and 4 different regions with discounts applied from 0% to 30%. Key features in those categories are as follows:

- Product Categories-Books, Electronics, Fashion, Beauty, Sports and Home & Kitchen
- Regions- North America, Europe, Asia and the Middle East
- Payment Method- Credit Card, Debit Card, Wallet, UPI and Delivery

Section 5. Visualization Tools

I chose to use Excel for all of my visualizations. With its extensive built in options for visuals it allowed me to complete all of my goals without transferring data between different tools. With the size of the dataset I also liked Excel for the pivot table option.

Section 6. Visualizations and Stories

Goal 1: Total Revenue by Product Category

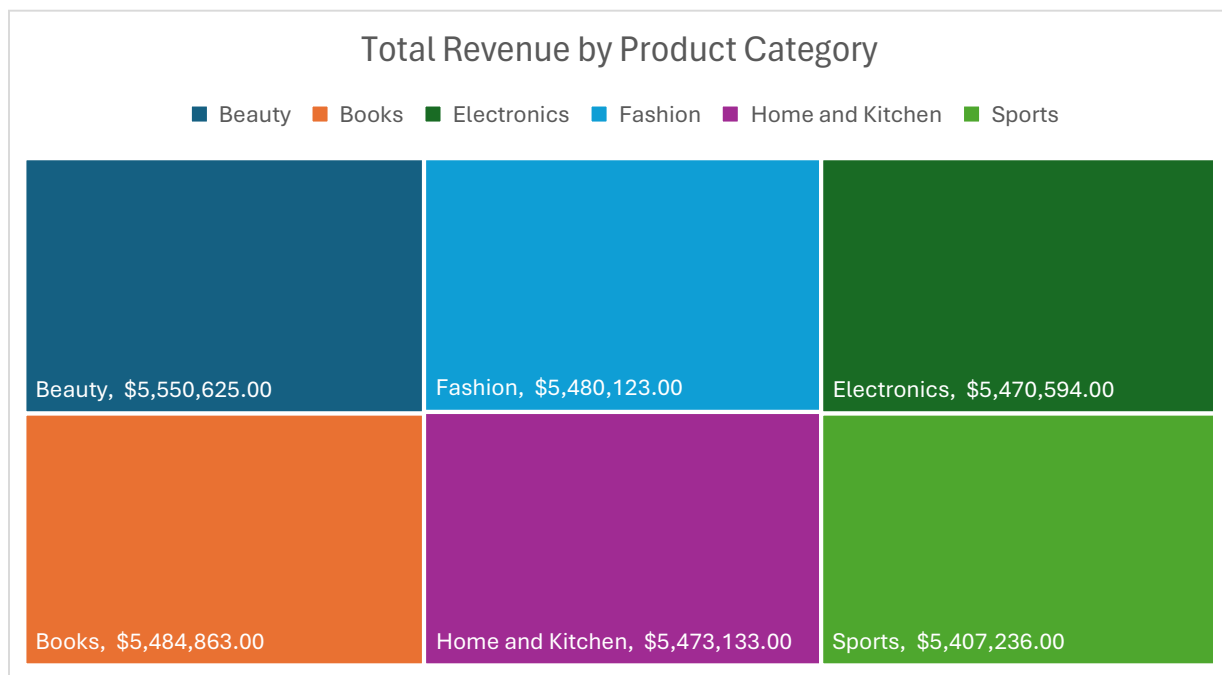
Goal: Does one product category generate more total revenue than the other five or is revenue equally disbursed across all categories?

Attributes:

Product Category- the category of the product purchased in the transaction

Total Revenue- sum of the total price of the transaction for all products purchased after the discount is applied for each of the 6 specific product categories

Selected Chart Type: Tree Map-The tree map allows for easy visual comparison of parts to a whole. In this specific case it will allow the user to review the total revenue of an individual product category like Books compared to the other categories and to the whole sum of the revenue for transactions in the dataset.



From the chart the total revenue from each of the 6 product categories is very evenly disbursed with Sports generating the lowest revenue totaling \$5,407,236.00 and Beauty generating the highest revenue at \$5,550,625.00. The difference between the 2 categories is just \$143,389 or less than half a percent of the total revenue of all the transactions combined. From this the Beauty product category could push additional banner ads for their products to site users or launch an email campaign advertising additional beauty products recommendations based on prior purchases to try and extend their revenue lead. Sports and Fashion could take a more seasonal approach, focusing their product advertisements at peak season like Sweaters in winter or swimming pools in summer.

Goal 2: Total Revenue by Product Category

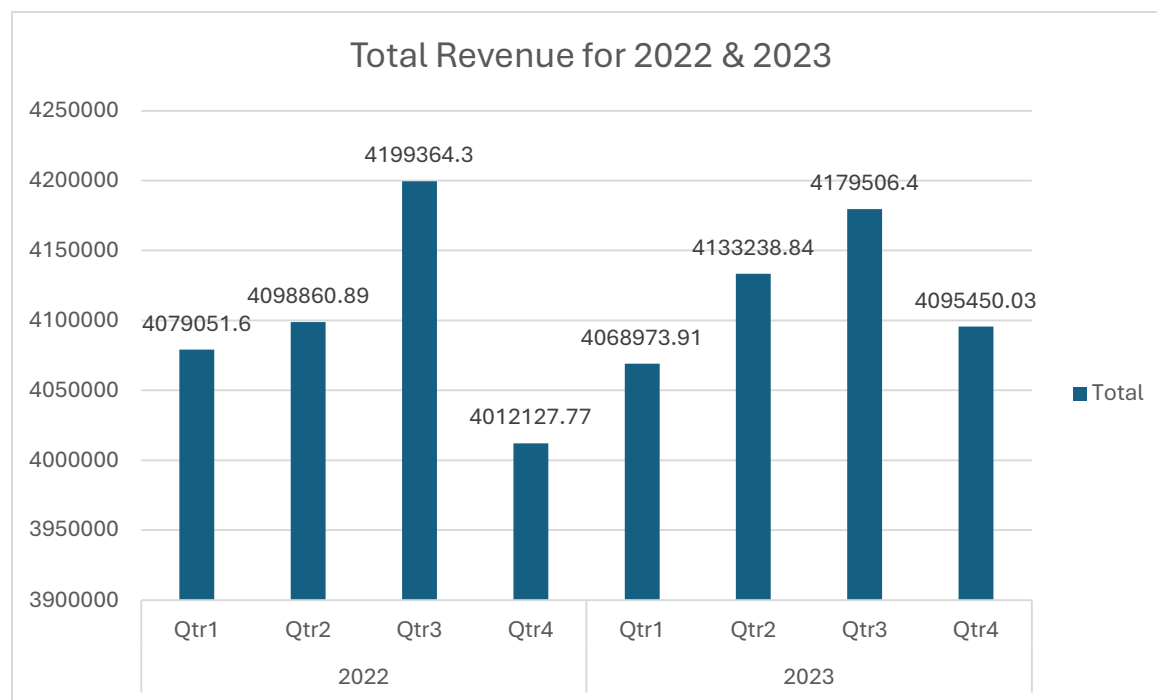
Goal: Does one quarter outperform others or is revenue generated evenly over all quarters? Is there any revenue growth year to year?

Attributes:

Order Date- the date the order was placed broken out by year, quarter and month.

Total Revenue- sum total price of the transaction for all products purchased after the discount is applied for the specific period of time.

Selected Chart Type: Clustered Column- The clustered column chart is idea for comparing multiple series of data side by side. Allowing for ease of use in identifying potential outliers or trends. Specifically in this chart we are comparing quarterly total revenue between 2022 and 2023.



From the chart you can see a fairly close distribution of revenue between the 2 years with 2023 generating just slightly more than 2022. You can also see a similar distribution of revenue between the quarters within the years with Quarter 2 generating more than Quarter 1, and Quarter 3 generating more than both Quarter 1 and 2 before Quarter 4 drops off. The major difference between the 2 years is there is a more dramatic spike in revenue in quarter 3 before a sharp fall in quarter 4. I would suggest, based off of these results that a greater marketing campaign be launched in quarter 4 offering larger discounts to help encourage more orders, especially around the holidays.

Goal 3: Total Revenue by Region

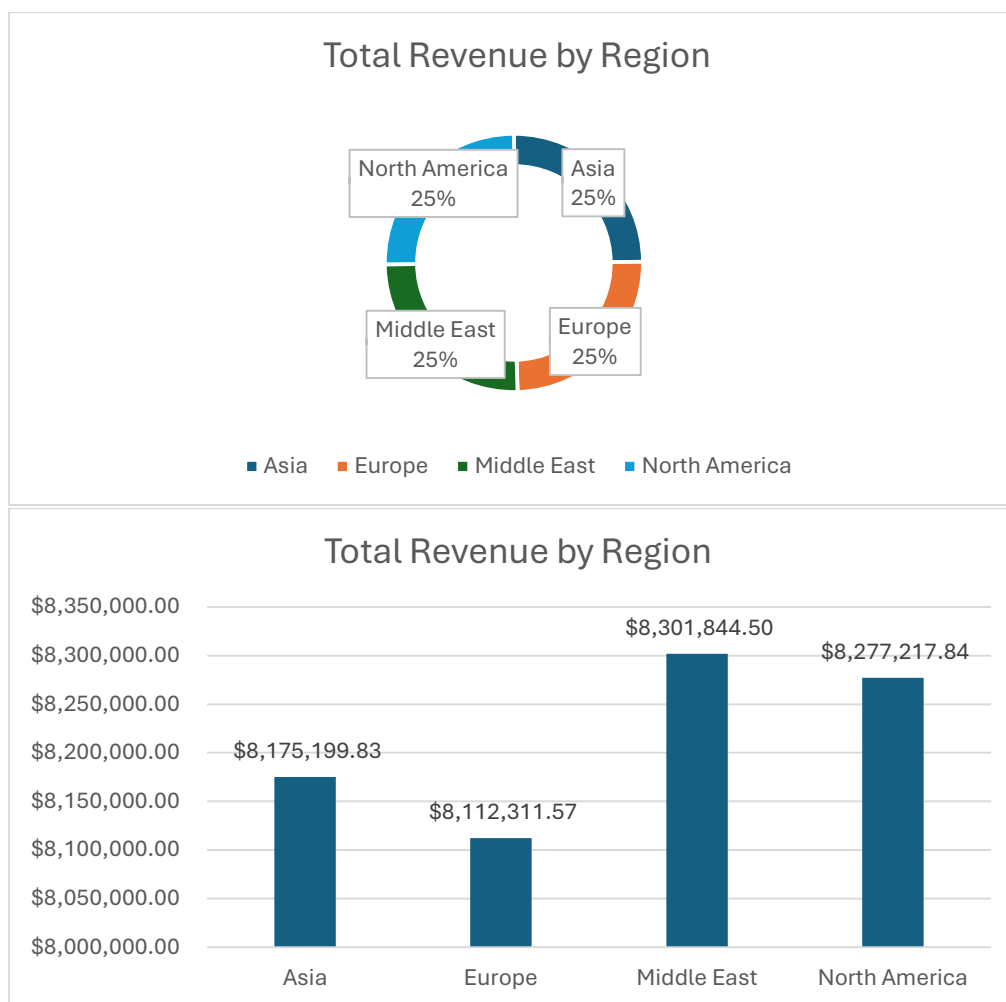
Goal: Which Region generates the most total revenue for both 2022 and 2023?

Attributes:

Region- Country/Continent the customer ordered in

Total Revenue- sum total price of the transaction for all products purchased after the discount is applied for the each region

Selected Chart Type: Bar Chart & Donut Chart- The bar chart is ideal for direct comparison of an attribute. While the donut chart is The donut chart is ideal for comparing parts to a whole. For both charts we compare total revenue in 2022 and 2023 across each of the 4 regions.



From the donut chart we can see that each of the regions makes up an equal share of the total revenue when expanding out and looking at 2 decimal points there is a small difference between the regions but nothing greater than about half a percentage point. When looking at the bar chart the differences between the revenue generated in the regions is more clearly visual. The Middle East generated the most revenue with \$8,301,844.50, while Europe generated the least with \$8,112,311.57. From these

results a marketing campaign focused in Middle East top performing product categories or a special discount offered to Middle East customers to encourage repeat business to generate more potential total revenue to help close the gap in the regions.

Goal 4: Average Total Revenue by Product Category

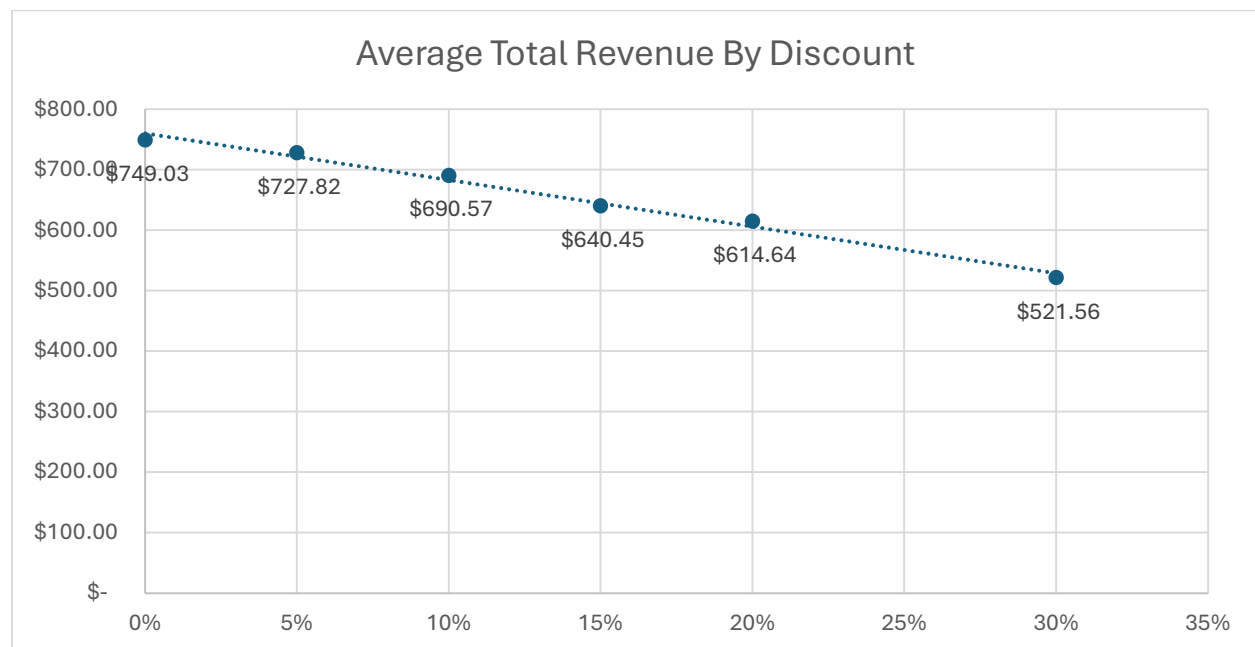
Goal: Do transactions with higher discounts encourage more spending?

Attributes:

Discount Percent- discount percentage applied to the transaction

Total Revenue- average total price of the transactions for all products purchased after a specific discount is applied

Selected Chart Type: Scatterplot- The scatterplot is ideal when trying to visualize a relationship between 2 variables and identify trends.



The average total revenue for a transaction with no discount applied is \$749.03; while the average for a transaction with a 30% discount is \$521.56. From the visual you can conclude there is a clear relationship between the discount percentage applied to the transaction and the total revenue generated. Unlike what was assumed where the larger the discount the more the customer would be encouraged to spend the trend is downward showing the higher the discount the lower the total revenue. It also appears that the as the discount percent increases the larger the decrease in total revenue is. The loss of revenue between 0% and 5% revenue is \$21.21 while 5% and 10% is \$37.25. From this I would recommend eliminating the 30% discount, or rolling out a 25% offering that could increase the average revenue for those transactions.

Goal 5: Payment Method by Region

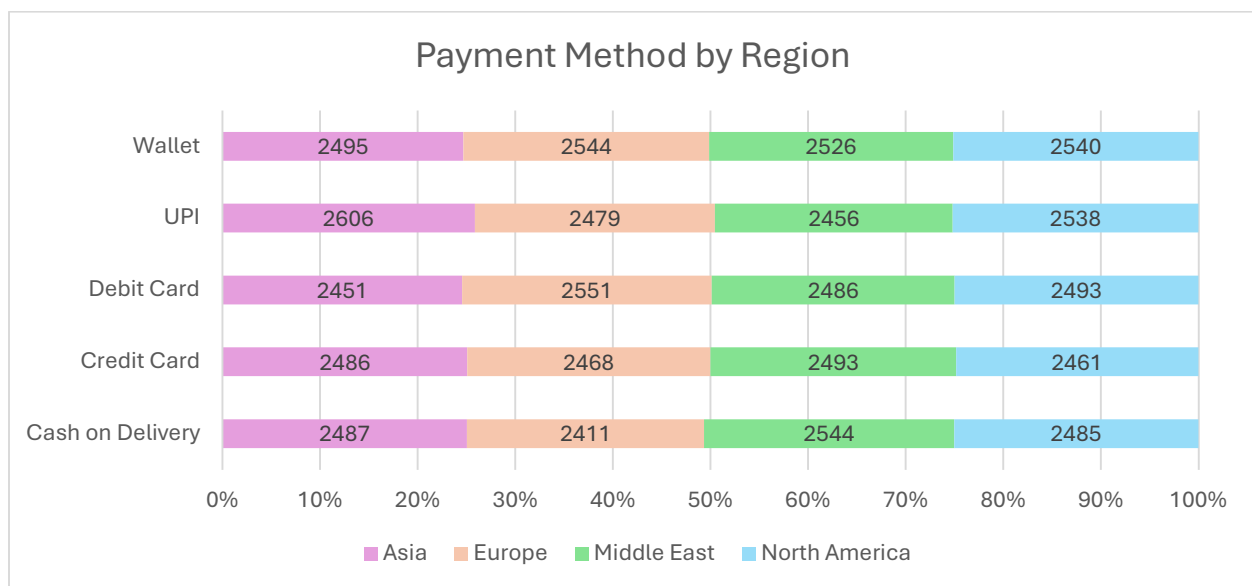
Goal: Does region affect the customers chosen payment method?

Attributes:

Region- Country/Continent the customer ordered in

Payment Method- sum total of number of transactions where a specific method of payment was used in a specific region

Selected Chart Type: Stacked Bar- The stacked bar chart allows for comparison of total values across groups while also being able to visualize the parts that make up the total.



The visual shows fairly even splits of payment methods across all regions. There is no clear outlier with significantly more of less chosen use than another. In the Asian region UPI completed transactions are coming out on top with 2606 orders; while Debit Cards come out on top in Europe with 2551 orders. In the Middle East Cash on Delivery is the leader with 2544 orders with Wallet leading North America with 2540 orders. From these results I would recommend offering a corresponding 5% discount with the leading payment methods across the regions to encourage more orders without taking a significant decrease in revenue. Or in the opposite the discount could be offered to customers using the least chosen payment method to help close the small gaps in each region.

Goal 6: Product Ratings by Category

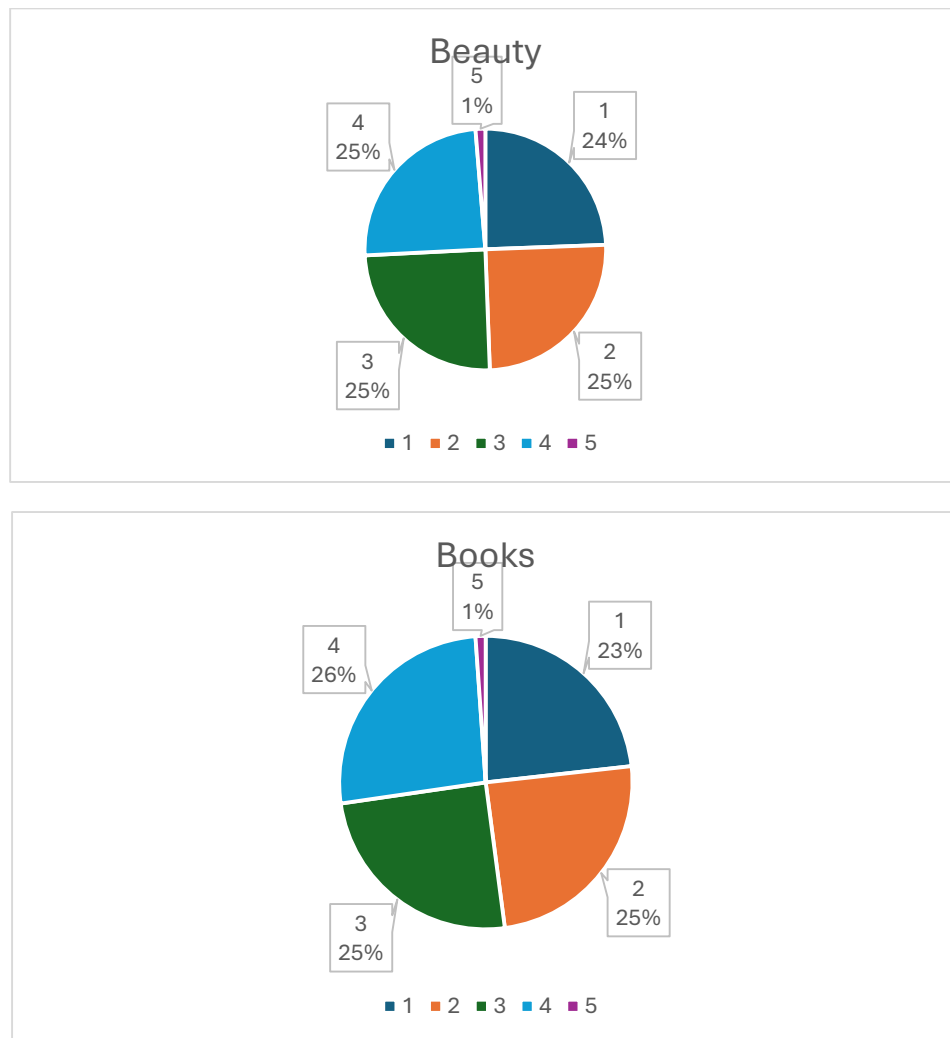
Goal: Do specific product categories generate better reviews than other?

Attributes:

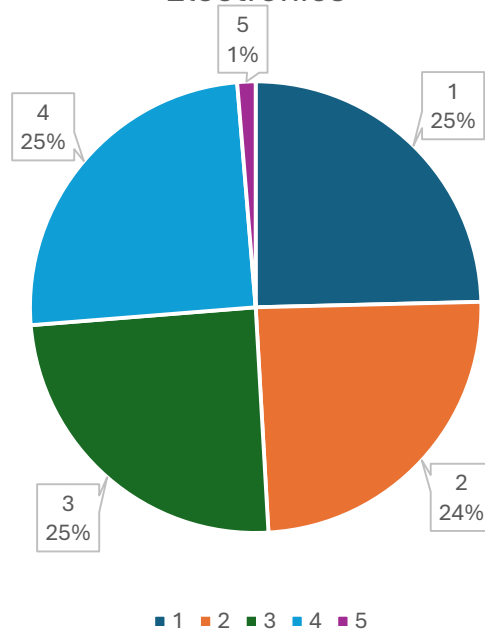
Product Category- the category of the product purchased in the transaction

Payment Method- sum total of the number of customer ratings for a specific product category broken out by scale 1 through 5

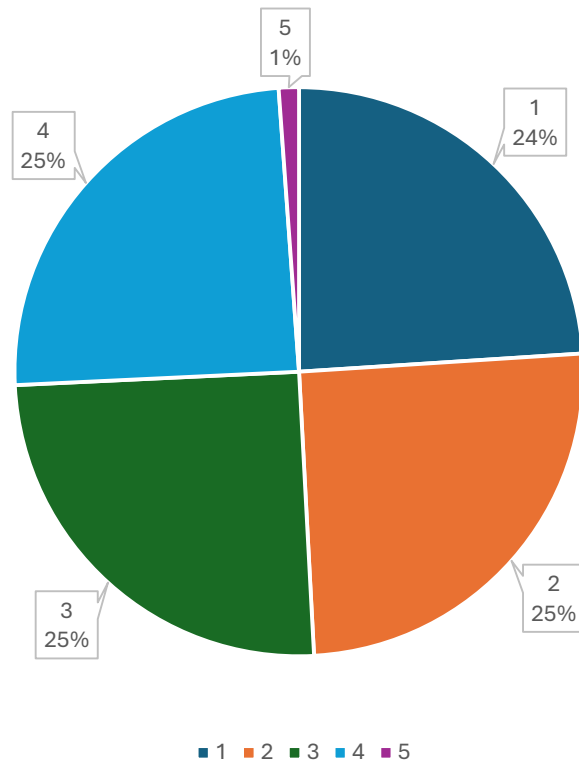
Selected Chart Type: Small Multiples- Small Multiples will allow for the comparison across product categories without overwhelming the audience with 1 dense multilayered chart allowing for the chance to highlight pattern and not just significant values.

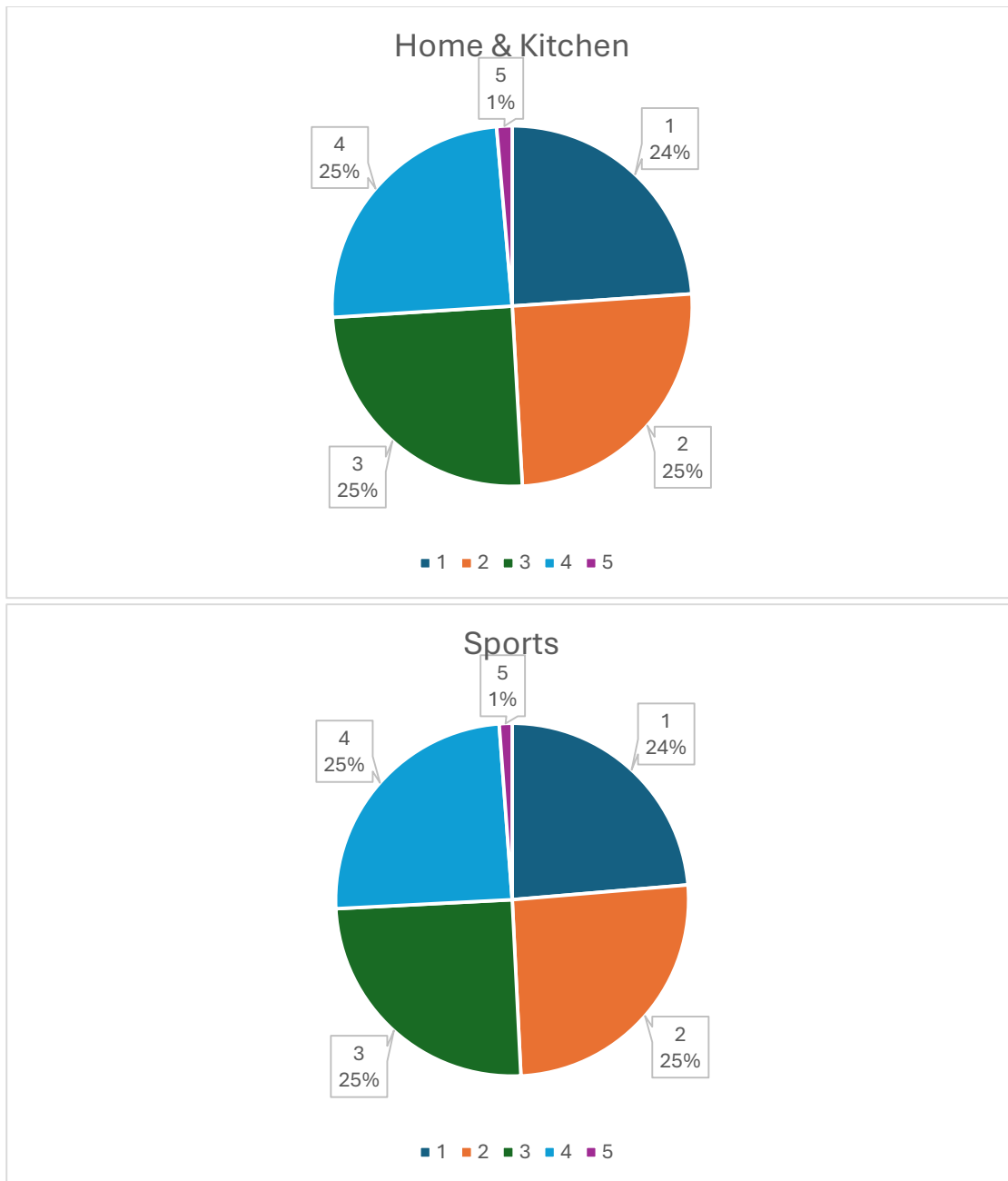


Electronics



Fashion





In all categories ratings are broken down very similar with no more than a percentage point difference in between categories. Books and Electronics bring in higher reviews on average while Beauty in Fashion average a lower rating overall. Overall the results aren't surprising because most people spend more time researching larger purchases like electronics before buying while beauty and fashion are dependent on trends and change seasonally. I would suggest a closer look at the reviews that scored between 1 and 1.9 to determine how to start increasing overall customer satisfaction.

Section 7. Conclusions

The project analyzed over 50,000 transactions from 2022 and 2023 and explored revenue distribution, seasonal trends, regional performance, discount effectiveness, payment method preferences, and product ratings. Using excel to generate visuals and draft narrative the following conclusions were made:

- Revenue was evenly distributed across product categories with Beauty generating the highest total and Sports the lowest which suggests for targeted marketing and email campaigns to push similar products to past customers of the product category.
- Regional performance by revenue was also very evenly distributed, with the Middle East leading with the highest total leading to recommending sales campaigns to Middle Eastern Customers to encourage repeat business and increase total revenue in the region
- Payment methods were fairly consistent across all regions with a different method leading in each region leading to sales campaigns tied with use of a specific payment method to encourage increased transactions.
- The higher the discount the lower the resulting average transaction total leading to suggest that they discontinue offering the 30% of discount and offering more at a lower rate between 5-15%
- Quarterly Total Revenue distributed similarly between 2022 and 2023 with 2023 generating more total revenue overall. This lead to suggesting more customers be offered discounts in quarter 4 to try to close the revenue gap.